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STATE OF THE ART REVIEW
Digital health and innovation

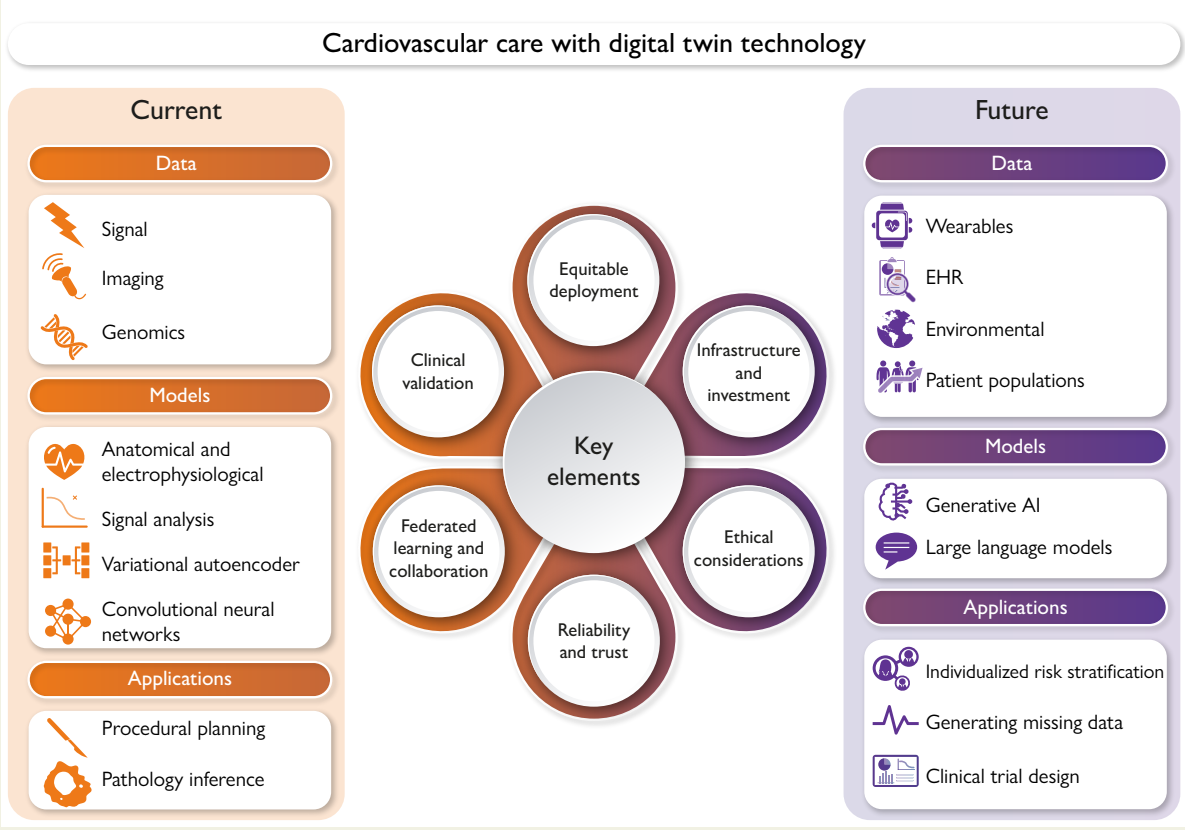
Cardiovascular care with digital twin technology in the era of generative artificial intelligence

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Graphical Abstract



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An overview of elements comprising current digital twins in cardiovascular medicine (left panel), elements needed to build future digital twins in cardiovascular medicine (right panel), and key elements for implementation of the digital twin in healthcare (center). AI, artificial intelligence; CABG, coronary artery bypass graft; EHR, electronic health record.

Abstract

Digital twins, which are *in silico* replications of an individual and its environment, have advanced clinical decision-making and prognostication in cardiovascular medicine. The technology enables personalized simulations of clinical scenarios, prediction of disease risk, and strategies for clinical trial augmentation. Current applications of cardiovascular digital twins have integrated multi-modal data into mechanistic and statistical models to build physiologically accurate cardiac replicas to enhance disease phenotyping, enrich diagnostic workflows, and optimize procedural planning. Digital twin technology is rapidly evolving in the setting of newly available data modalities and advances in generative artificial intelligence, enabling dynamic and comprehensive simulations unique to an individual. These twins fuse physiologic, environmental, and healthcare data into machine learning and generative models to build real-time patient predictions that can model interactions with the clinical environment to accelerate personalized patient care. This review summarizes digital twins in cardiovascular medicine and their potential future applications by incorporating new personalized data modalities. It examines the technical advances in deep learning and generative artificial intelligence that broaden the scope and predictive power of digital twins. Finally, it highlights the individual and societal challenges as well as ethical considerations that are essential to realizing the future vision of incorporating cardiology digital twins into personalized cardiovascular care.

Keywords Digital twins • Generative artificial intelligence • Multi-modal models • Precision medicine

Introduction

Digital twins in the era of generative artificial intelligence (AI) can enable true precision cardiovascular care.[1,2] In *Figure 1*, we present a clinical scenario that envisions digital twin-enabled precision care. This contrasts with current practice, where most care decisions are driven by clinicians trialling different therapies without information on the expected therapeutic response for individuals and requiring frequent in-person evaluations with static instruments that are inconvenient for both the patient and clinician.[3,4] Digital twins are dynamic virtual patient representations that can personalize care by integrating multi-modal data from the patient and simulating clinical scenarios and treatment strategies to forecast outcomes, health status changes, and solutions. These data include dynamic features tracked by environmental sensors, wearable fitness trackers, and at-home health monitors, along with immutable and historical health data such as genomics and medical records.[5] The integration of these data into deep learning models developed in broad and diverse patient populations enables the comparison of similar care scenarios to define accurate and personalized predictions, which are updated in real-time with feedback. There is unrealized potential for digital twin technology in cardiovascular care, which has been limited to disease phenotyping and broad treatment strategy definitions, with a future role that provides a holistic and dynamic record and guide for a patient's lifetime.

What are digital twins?

Digital twins are *in silico* replicates of a subject, utilizing multi-modal real-time data to build models optimizing decision-making and forecasting an intervention's influence on outcomes.[6,7] These replicates build accurate simulations while simultaneously receiving feedback through a continuous connection with the subject, known as the cyber-physical space.[6] The concept originated in the aerospace industry and has established utility in production workflows in industrial manufacturing and traffic management in smart cities, among other applications.[2,6] The data representing the inputs for these digital twins vary in granularity, and the outputs are connected with the digital twin through wireless communication, enabling real-time feedback and adjustments.[6] These digital twins are generally powered by mechanistic models that use established scientific principles in mechanical engineering, fluid dynamics, and electromagnetism more than data-driven factors.

Although the digital twin is the established state-of-the-art simulation architecture for engineering, it is still a developing field in healthcare. There are varying definitions of a healthcare digital twin, with some restricted to 'passive' data collection, wherein future health states are predicted based on models that leverage large scans of retrospective data. In contrast, 'active' feedback from the patient utilizes continuous, real-time data streams that reactively update predictions.[8] The latter has been challenging because patient data are variably acquired, mechanistic models of disease pathophysiology are incomplete, and factors outside biological constraints, such as human behaviour that are not captured as inputs, influence the success of interventions. This contrasts domains such as aviation, which have set design and safety principles.[9] For example, outcomes of cardiovascular interventions established through randomized clinical trials (RCTs) have not always had a clear biological explanation, such as the mortality benefit from sodium–glucose cotransporter 2 (SGLT2) inhibitors or the higher mortality with intensive blood glucose-lowering or with prescribing anti-arrhythmic medications after myocardial infarction.[10–12] New advances in data acquisition, integration of real-time and static patient data in deep learning models, and evidence generation and translation through generative AI provide opportunities to broaden the scope of cardiovascular digital twins (*Figure 2*).

Digital twins fill several important gaps. First, simulating several procedural approaches in one digital twin allows a safe and optimized procedure on the actual patient. It also allows the examination of interacting elements, such as an individual's environment and genome, which might influence their outcome. In addition, it is infeasible to run RCTs necessary to test all permutations of an intervention, but digital twins can model these scenarios across broad populations with specific controls to address confounding. The eventual application is a pipeline that automates the analysis of multi-modal data and integrates them into clinical decision-making.[13] Finally, the feedback from the success or failure of an intervention will improve the model performance compared with clinicians, whose diagnostic heuristics may affect decision-making.



Ms. K is a 76-year-old woman with heart failure with preserved ejection fraction (HFpEF), type 2 diabetes, hypertension, and obesity and is currently walking outside.

Her smartwatch detects and records in her medical digital twin:

- *A 6-second episode of atrial fibrillation*
- *A modest decrease in oxygen saturation*
- *An outdoor temperature of 35ºC with 70% humidity, Air Quality Index: 125*

Her wireless blood pressure cuff records in her digital twin: 170/93



Her arrhythmia and elevated blood pressure have triggered her EHR to run an LLM extracting:

- *Relevant medical history, vital signs, ECGs, cardiac imaging, medications*
- *A previous echocardiogram report noting: "diastolic dysfunction" and an "enlarged left atrium"*
- *A recent clinic note recommending: "Pause HCTZ due to a decrease in potassium and rise in uric acid."*



The digital twin model has determined: Ms. K is likely to experience a heart failure exacerbation. It then:

- *Runs simulations of different blood pressure and diuretic medications on her outcome*
- *Compares this patient's digital twin to others who are similar in profile*
- *Draws on its knowledge of relevant randomized clinical trials and guidelines.*



The digital twin Ms. K's physician with a recommendation:

- *Undergo a basic metabolic panel*
- *Restart HCTZ, initiate low-dose furosamide, and start an SGLT2 inhibitor and spironolactone*

Ms. K's physician then verifies the recommendation summary and contacts Ms. K with a plan. The model finally updates its training with this episode to inform future predictions.

Figure 1 A clinical vignette. A step-by-step clinical vignette modelling the future of healthcare with digital twins. HCTZ, hydrochlorothiazide; SGLT2, sodium–glucose cotransporter 2

In this review, we summarize the current role of digital twins in cardiovascular care and introduce advances in data acquisition and AI that further enable their capacity, discuss challenges in their implementation, and forecast future directions to achieve their immense potential (*Graphical Abstract*).

Digital twins in cardiology

Advancing precision medicine through digital twins spans deep phenotyping of complex cardiac disorders and individualizing treatment, such as coronary and structural interventions as well as electrophysiological procedures (*Table 1*).[14–46] Below, we summarize these innovations, while also discussing the uses of digital twins combined with generative AI for generating synthetic data to improve the understanding of pathophysiology and conducting *in silico* clinical trials.

Digital twins for precision medicine in cardiology

Digital twins in cardiomyopathies

Digital twins enable granular phenotyping for heterogeneous disorders such as cardiomyopathies characterized by complex myocardial, mechanical, and electrical remodelling patterns.[47] These distinct signatures have been detected on a myriad of data streams[48,49] and increasingly have implications for the diagnosis, prognostication, and management of these disorders.[49–52] They may also underlie some of the treatment effect heterogeneity described for various commonly used therapeutic approaches, such as those in the setting of cardiac resynchronization therapy (CRT),[53] implantable cardioverter defibrillators,[54] and disease-modifying therapeutics.[55–57] The multimodal assessment of these patients, which generally encompasses 12-lead electrocardiography (ECG) along with echocardiography and cardiac magnetic resonance (CMR), further enables a granular comparative assessment of different patients and therapies.[47,58]

In recent years, the integration of geometrical and topological features from cardiac imaging and ECG has enabled personalized, *in silico* simulation of electromechanical coupling and its haemodynamic implications from abnormal conduction rhythms such as left bundle branch block (LBBB).[59,60] Digital twin models have also been used to enhance disease-specific predictive models, such as stratifying the risk of ventricular arrhythmias in ischaemic cardiomyopathy by incorporating dynamic aspects of myocardial physiology and evolving anatomical substrates, triggers, and modulators.[61] In addition, in a small-scale retrospective analysis of 45 patients with cardiac sarcoidosis, personalized fusion models of CMR and positron emission tomography improved the prediction of future arrhythmias.[62]

Digital twins may also guide more precise phenotyping of rare forms of cardiomyopathy. In an analysis of 313 patients with arrhythmogenic right ventricular cardiomyopathy (ARVC), echocardiography-based digital twin models linked deformation abnormalities to the underlying myocardial disease substrate, enabling deeper insights into the mechanical underpinnings of progression and eventual prediction of ventricular arrhythmia events.[63] In a separate study of patients with ARVC, a genotype-specific digital twin approach (geno-digital twin) combined genotype-derived cellular electrophysiological simulations with CMR to accurately predict the location and mechanism of ventricular tachycardia circuits.[64] This paradigm has recently been extended to congenital heart diseases, where information from computed tomography (CT) scans and ECGs are integrated into a computational approach defining the electrophysiological activation and repolarization patterns in a patient with hypoplastic left heart syndrome.[65]

Coronary and structural heart simulations

Digital twins simulate procedural decision-making on patient physiology, ensuring a higher likelihood of success without invasive

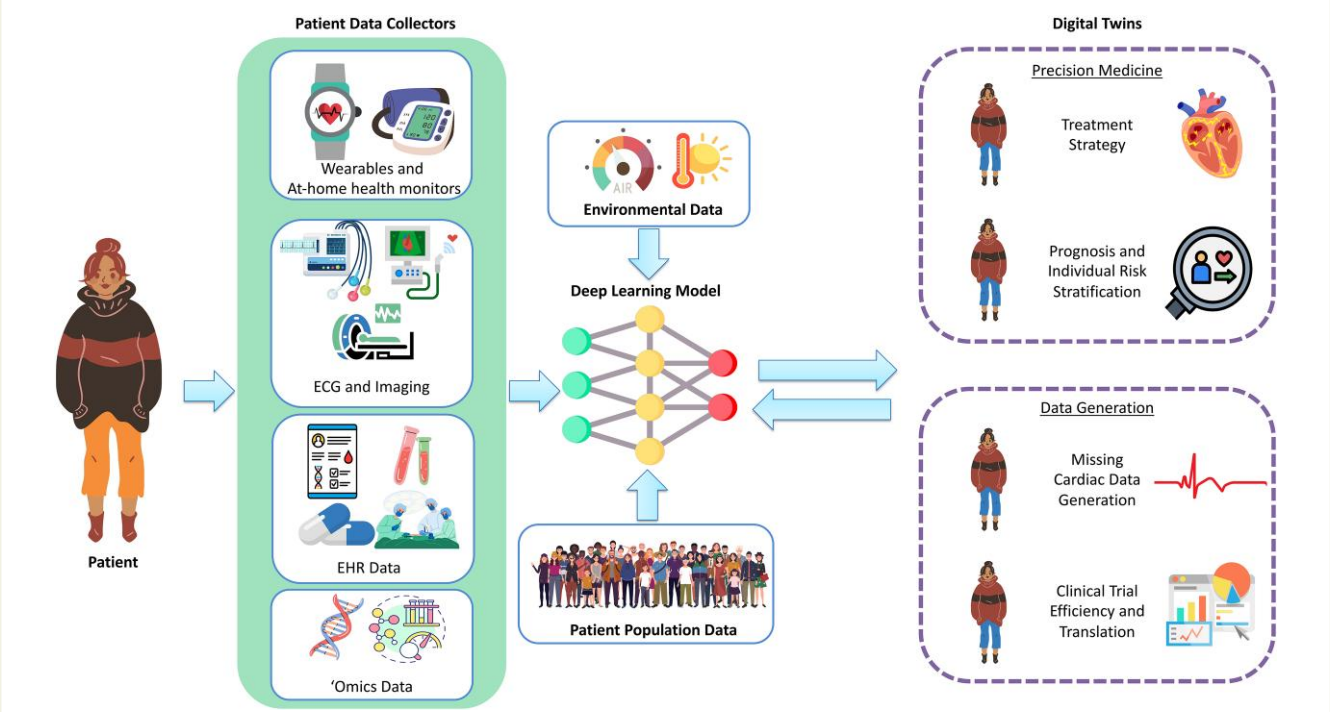


Figure 2 Potential cardiovascular clinical applications of digital twins. An overview of data available and potential digital twin applications. ECG, electrocardiogram; EHR, electronic health record, 'Omics: includes genomics, proteomics, and metabolomics data

experimentation (*Figure 3*). For example, coronary CT angiography with fractional flow reserve (CT-FFR) is an imaging study that builds 3D reconstructions of atherosclerotic coronary vessels, replacing the need for invasive measurements when planning for percutaneous coronary interventions. Digital twins constructed from CT-FFR have combined reconstruction with haemodynamic modelling to accurately predict the degree of coronary artery stenosis and enable virtual coronary intervention simulations.[21,22]

In addition, applying computational fluid dynamics and wall shear calculations to these CT-FFR digital twins can optimize the selection of bypass grafts with higher accuracy in predicting 1-year patency than selection without assistance from this technology.[15,18,24] A pilot study of 114 patients, FASTTRACK CABG, is testing the use of CT-FFR digital twins for surgical planning for coronary artery bypass graft.[66] In preventive cardiovascular care, the CREDENCE study of 303 patients combined CT-FFR with neural network models to predict atherosclerosis progression, with an accuracy of 84% for ≥50% stenosis prediction and a correlation of .73 with quantitative coronary angiography.[67]

Outside of coronary procedures, digital twin models for structural heart interventions, such as valve replacement, can align anatomical structures to prosthetic devices. A pair of proof-of-concept studies with 9 and 80 patients, respectively, undergoing transcatheter aortic valve replacement (TAVR) showed that CT-derived computer simulations could improve TAVR depth implantation estimates to reduce post-procedural paravalvular regurgitation.[25,26]

Electrophysiology and electromechanical models

Digital twins can also combine cardiac electrophysiology with cardiac electromechanics to advance procedural planning to treat cardiac arrhythmias. For example, simulating the physiological effects of CRT in the presence of LBBB[60] can guide appropriate lead placement. In an analysis of 45 patients with heart failure referred for CRT, virtual pacing simulations using echocardiography-based digital twins predicted subsequent ventricular remodelling based on a reduction in the *in silico* septal-to-lateral work imbalance.[68] Similar concepts are now increasingly used across all electrophysiological procedures, including the ablation of ventricular tachycardia and atrial fibrillation (AF)[46,69,70] and identifying areas for radiofrequency ablation.[29,34] Beyond predicting treatment response and prognosis, integrating ventricular anatomical information by CMR and electrophysiological function information based on a 12-lead ECG can simulate human physiology variation across a patient population,[43] paving the path towards low-cost *in silico* clinical trials.

In one example, researchers integrated CMR and ECG recordings across two patients with hypertrophic cardiomyopathy and four controls to simulate individualized regional electromechanical coupling effects.[45] In another application, a virtual assessment of anti-arrhythmic drug efficacy was simulated in a single-centre study of 232 individuals at high risk of recurrent AF following catheter ablation. By integrating CT and electroanatomical maps and knowledge of the pharmacological properties of distinct anti-arrhythmic drugs, the treatment that virtually terminated AF the fastest was associated with a lower risk of AF recurrence during follow-up relative to alternative agents.[71]

Data, models, and applications to advance digital twins in cardiology

As described above, current digital twin approaches in cardiovascular medicine generally use multi-modal data to define a phenotype with

Table 1 Summary of electrophysiological and electromechanical models

Year	Study application	Data	Model type	Validation size	Simulation use	Authors
2013	HF	CMR/ECG	EM	Individual ($n = 11$)	Regional contractility LV	Marchesseau et al.[14]
2016	CABG planning	CTA	CFD	Individual ($n = 5$)	Coronary HD	Ramachandra et al.[15]
2016	Post-MI arrhythmia	CMR	EPM	Individual ($n = 41$)	SCD risk	Arevalo et al.[16]
2016	Healthy	CMR	EPM	Individual ($n = 3$)	AP, MD, and TS of HB	Augustin et al.[17]
2017	CABG planning	Ang./CT	CFD	Individual ($n = 10$)	Coronary HD	Ballarin et al.[18]
2017	Healthy	Simulated data	EPM	Simulated data	AP, HD, MD, and TS of HB	Kosta et al.[19]
2018	Post-MI arrhythmia	CT	EP	Individual ($n = 7$)	VT ablation planning	Cedilnik et al.[20]
2019	PCI planning	Ang.	CFD	Individual ($n = 54$)	Post-PCI FFR	Gosling et al.[21]
2019	PCI planning	CCTA/Ang.	CFD	Individual ($n = 24$)	Post-PCI FFR	Modi et al.[22]
2019	AF	CMR	EPM	Individual ($n = 10$)	AF ablation planning	Boyle et al.[23]
2020	CABG planning	CCTA	CFD	Cohort ($n = 12$)	Graft patency	Zhu et al.[24]
2020	TAVR planning	CT	EM/FEM	Cohort ($n = 9$)	PVL	Dowling et al.[25]
2020	TAVR planning	CT	EM	Cohort ($n = 80$)	PVL	El Faquir et al.[26]
2020	Healthy	CMR	GP/FEM	Individual ($n = 1$)	AP of Atrium	Hu et al.[27]
2021	Healthy	MRI/ECG	EP	Individual ($n = 1$)	12-lead ECG	Gillette et al.[28]
2021	Healthy/scar	CMR	EPM	Individual ($n = 1$)	Ablation scar modelling	Gerach et al.[29]
2022	PCI	OCT	FEM	Individual ($n = 2$)	LA	Poletti et al.[30]
2022	Healthy	CMR	EPM	Individual ($n = 1$)	PV, TS, strain	Stimm et al.[31]
2022	Healthy	ECG/CMR	VAE	Cohort ($n = 30$)	Ventricular activation	Li et al.[32]
2022	Aortic coarctation	CMR	EM	Individual ($n = 7$)	AP, MD, TS of HB	Jung et al.[33]
2023	Healthy	N/A	FEM/EM	N/A	AP, MD, TS of HB	Fedele et al.[34]
2023	Healthy	CMR	Hybrid GAN/U-net	Individual ($n = 18$)	Synthetic CMR	Xing et al.[35]
2023	Healthy	CT	VAE/diffusion	Cohort ($n = 512$)	Volume	Kadry et al.[36]
2023	Post-MI VT	CMR	EP	Individual ($n = 21$)	VT risk	Serra et al.[37]
2023	HF with AF undergoing CRT	ECG-gated CT, IPM	EM	Individual ($n = 1$)	AP, HD, MD, TS of HB	Strocchi et al.[38]
2023	Healthy	CMR/CT	SSM	Individual ($n = 1$)	Cardiac anatomy	Azzolin et al.[39]
2023	Healthy	OFI	Hybrid CNN/PINN	Simulated/*ex vivo*	AP	Kashtanova et al.[40]
2024	Healthy	Simulated data	PINN	Simulated data	AP	Jiang et al.[41]
2024	MI scar from healthy	ECG/CMR	EP/VAE	Individual ($n = 1$)	Simulate MI scar	Li et al.[42]
2024	Healthy	ECG/CMR	EP	Individual ($n = 1$)	Dofetilide dosing	Camps et al.[43]
2024	LVAD/MS	ECG/US	PINN	Cohort ($n = 200$)	PV loop	Kuang et al.[44]
2024	HCM	ECG/CMR	FEM/EP/CNN	Individual ($n = 6$)	Purkinje network	Camps et al.[45]
2024	AF	CMR	EA	Individual ($n = 26$)	PV isolation ablation	Sakata et al.[46]

The first column describes the year of publication, the second column describes the study application, the third column describes the data used, the fourth column describes the type of model used, the fifth column describes the size and type of validation of the model, the sixth column describes the digital twin simulation use, and the final column is the publication reference.
AF, atrial fibrillation; Ang., angiography; AP, action potential; CCTA, coronary computed tomography angiography; CFD, computational fluid dynamics; CMR, cardiac magnetic resonance; CNN, convolutional neural networks; CT, computed tomography; EA, electroanatomical; ECG, electrocardiogram; EM, electromechanical; EP, electrophysiological; EPM, electrophysiological and electromechanical; FEM, finite element method; FFR, fractional flow reserve; GP, Gaussian process; HB, heartbeat; HD, haemodynamics; HF, heart failure; IPM, invasive pressure measurements; LA, lumen area; LV, left ventricular; LVAD, left ventricular assist device; MD, mechanical deformation; MI, myocardial infarction; MS, mitral stenosis; OCT, optical coherence tomography; OFI, optical fluorescence imaging; PINN, physics-informed neural network; PV, pressure volume; PVL, paravalvular leak; SCD, sudden cardiac death; TS, tissue structure; US, ultrasound; VAE, variational autoencoder; VT, ventricular tachycardia.

specified cardiac physiologic parameters to advance treatment strategy selection and prognostication of outcomes. These twins rarely have any dynamic connection to patients with the responsiveness to changing inputs and enable broader innovations in care. Digital twin models must adapt to external factors affecting the patient to predict treatment effects accurately. Specifically, with the new data and model

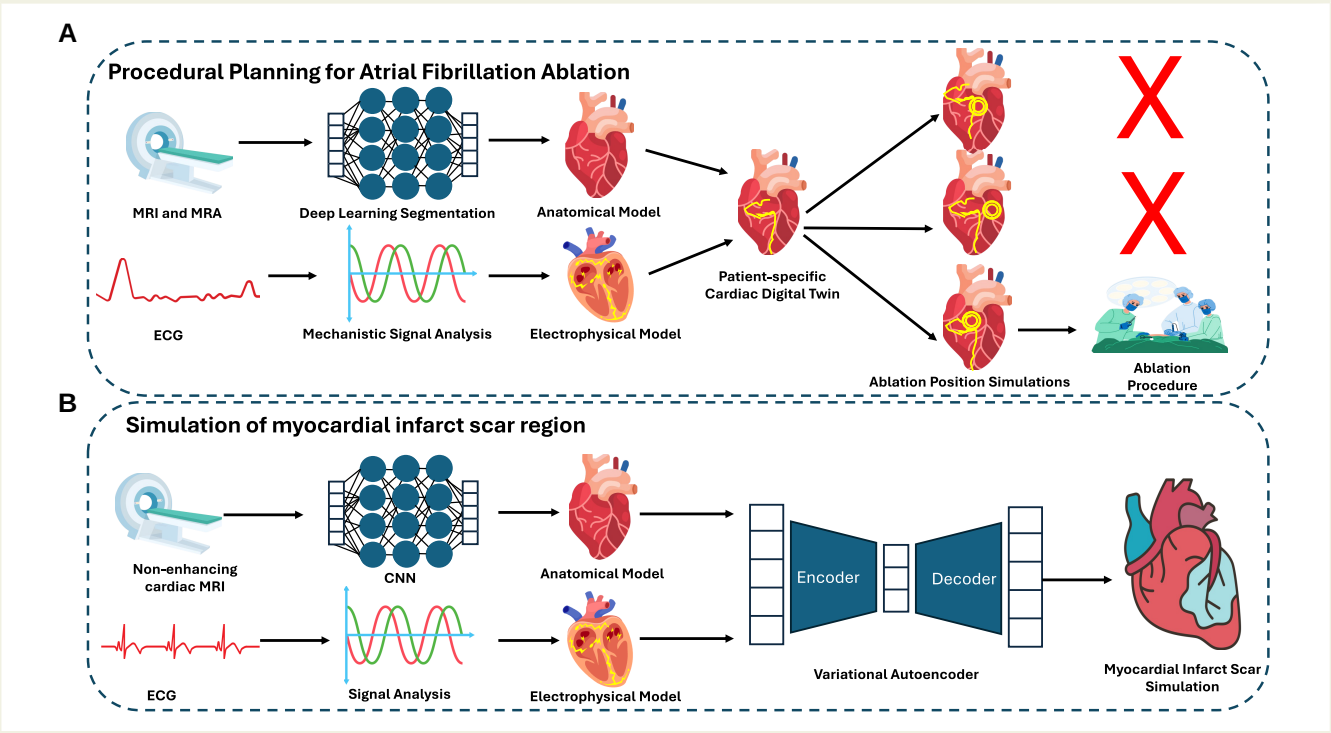


Figure 3 Selected precision medicine applications of digital twins. (*A*) An example digital twins for procedural planning of atrial fibrillation ablation. 3D anatomical and electrophysiological models are derived from magnetic resonance imaging, magnetic resonance angiography imaging, and electrocardiograms to build a patient-specific digital twin. Different ablation locations are simulated, and the one most likely to stop atrial fibrillation is chosen. (*B*) An example simulation of a myocardial infarct scar. To substitute late-gadolinium enhancement magnetic resonance images, non-contrast magnetic resonance images and electrocardiograms can be combined to make an anatomical and electrophysiological cardiac digital twin. The digital twin can then be input into a variational autoencoder, which can generate simulations of 3D scarring, e.g. from myocardial infarction. ECG, electrocardiogram; CNN, convolutional neural network; MRA, magnetic resonance angiography; MRI, magnetic resonance imaging

advances summarized below, digital twins in cardiovascular medicine can advance the granularity of phenotypes, predict outcomes with dynamic adaptation, and build novel *in silico* clinical trials.

Contemporary digital health data to augment digital twins

An ideal medical digital twin incorporates information from the cellular to physiological level to build a genomic, biological, mechanical, and environmental representation of a patient. These data, along with a continuous connection to the patient, comprise the medical digital twin cyber-physical space.[8] Population health data expand digital twins in cardiology from procedural planning to personalized prevention and intervention strategies in real-time. These include data in various historical formats, including tabular data from electronic health records (EHRs), imaging, ECGs, and genomics, and now dynamically available data from wearables and at-home health monitors used by the public. In addition, incorporating longitudinal data models changes a patient's phenotype from interactions with external factors, such as the environment and medical interventions. This section discusses the data types and patient cohorts available to advance the cardiovascular digital twin.

Historical data that builds a comprehensive but static patient health profile can be found in the EHR, which provides a large, high-dimensional dataset of clinical information essential for building medical digital twins. These include demographics, physiological biomarkers such as vital signs and laboratory values, and administrative codes for medical conditions, procedures, and medications, and also unstructured components such as medical notes.[72] The data are in chronological order and documented at varying time points. They also include digital data streams, such as ECGs; imaging, such as CT scans, magnetic resonance images (MRIs), and pathology images; and videos, such as point-of-care ultrasounds, echocardiograms, and cardiac catheterization.

In an attempt to make such multi-modal data part of characterizing individuals, large registries such as the UK Biobank and All of Us, have compiled some of these data by protocolized testing of a large number of individuals,[73] therefore, enabling the compilation of EHR data with cellular omics data, whole-body CTs, and MRIs.[74,75]

None of these data sources are set up for a real-time connection between the patient and the digital twin. Continuous monitoring of vital signs, blood-based biomarkers, and cardiac rhythms is exclusively seen in critically ill patients in the intensive care unit and relies on special equipment and personnel gathering these data.[76] Although this exemplifies the ideal data acquisition frequency, it limits the scalability and generalizability of digital twins of real-world patients. Wearable devices such as smartwatches, fitness trackers, continuous glucose monitors, and Bluetooth blood pressure monitors can complete the cyber-physical feedback loop with continuous metrics such as vital signs, arrhythmia screening, activity level, physical location, and environmental factors.[77]

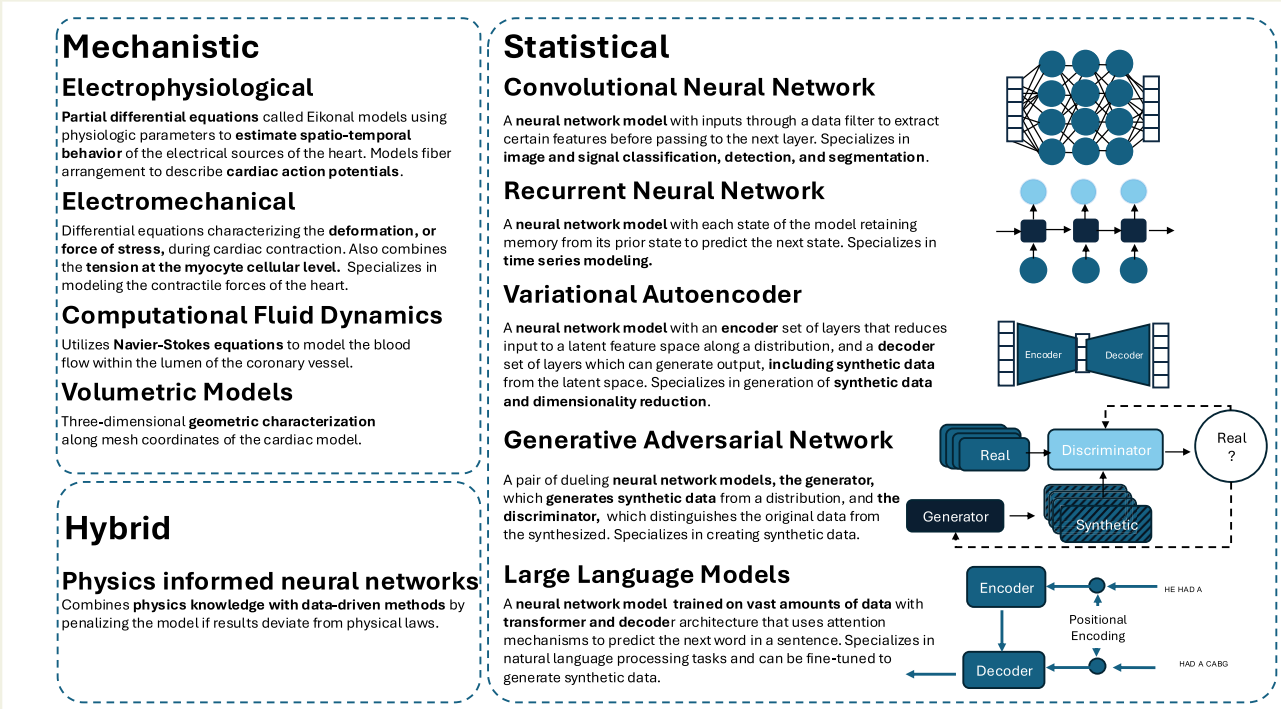


Figure 4 Methodological strategies for developing digital twins. An overview of mechanistic and statistical models used in cardiac digital twins

Methodological basis for cardiovascular digital twins

Mechanistic modelling strategies to build digital twins

Digital twins employ computational models that vary from mechanistic to statistical, depending on data availability and the purpose of the model (*Figure 4*). Mechanistic models employ physiologic parameters and principles of fluid dynamics, electrophysiology, and electromagnetism to characterize cardiac physiology and to simulate interventions to correct cardiac pathophysiology.

Statistical modelling strategies to build digital twins

Statistical models utilize probabilistic methods, including deep learning, to integrate high-dimensional multi-modal data and identify patterns across patient populations with the goal of predicting a wide range of cardiac pathology and processing longitudinal information.[8,78] Beyond establishing mechanistic and statistical models for digital twins, hybrid twins utilize physiological parameters while integrating lightweight deep learning models to process multi-modal data.

A growing number of deep learning-based digital twins are emerging to reduce computational complexity for applications without well-defined pathophysiology and for gaps in evidence.[35,41,42,79] The architecture of these deep learning models depends on the type of data (*Figure 3*). Current algorithms project multiple data sources to a shared embedding, a mathematical representation of the similarities and dissimilarities between different data types, such as ECGs, echocardiograms, and tabular data. This construct forms the basis of subsequent tasks and can be used to predict how changes in any of the input features may impact downstream events.[80]

Digital twins powered by generative AI models can advance the practice and understanding of cardiology through data generation. The most common deep learning method to generate digital twins of health data is a generative adversarial network (GAN), a deep learning model that comprises competing neural networks originally proposed for generating synthetic images.[81–86] These models discover complex hidden correlations between features. Specialized recurrent neural network models called transformers are adept at preserving the temporal structure of data and have been applied to EHR-based digital twins.[87]

Cardiovascular digital twins are set to be further enhanced by large language models (LLMs) that can expand patient representations from unstructured data in medical notes through inferences found from conditions, social determinants of health, or patient-reported medical history. Exemplified by GPT-4 (the model behind the popular ChatGPT), LLMs are adept at processing vast amounts of textual data, including medical records and research papers, to extract valuable insights and aid in diagnosis.[88]

Once the digital twin has been generated, it enables simulations to guide care. Beyond treatment strategy, digital twins can enhance outcome prediction. The integration of continuous data sources can update predictions in real-time while also giving feedback to develop the model for future simulations. Advances in deep learning have demonstrated accurate screening of cardiac disease normally diagnosed on echocardiograms using electrocardiograms and one-lead ECGs on wearables and detecting coronary artery disease from facial features.[89–92] This technology has also been incorporated into stethoscopes for screening at the point of care.[93]

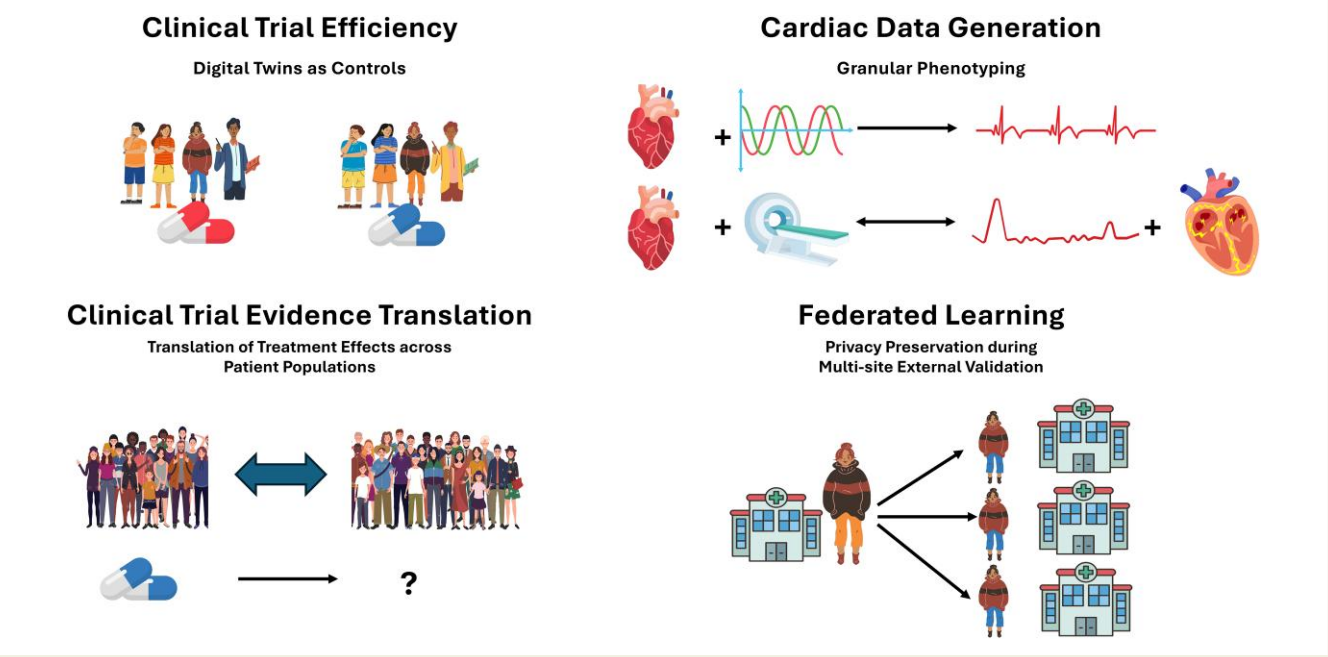


Figure 5 Applications of digital twins for evidence generation. An overview of digital twin applications for evidence generation across clinical trial populations, for generating cardiac data, for estimating treatment effects of a clinical trial in a second patient population, and for safely testing machine learning models across multiple sites

Digital twins for data generation: synthetic cardiac data

Often, digital twins are created from a select few data types available to gain further insights into the missing data and the cardiac function and pathology. Electrocardiography models, based on computational simulations and patient-specific data, such as MRIs, simulate the behaviour of the heart to produce realistic ECG waveforms.[28] By using anatomical information with known formulae based on the laws of electromagnetism, Gillette *et al.* have demonstrated the creation of synthetic ECG waveforms.[28] The direct link between the design principles from the mechanistic modelling of cardiac anatomy and the ECG waveform has many potential implications, such as the ability to generate large datasets of synthetic ECGs, potentially out of distribution, representative of different populations for training models.[94]

This will be especially valuable for those cardiomyopathies such as hypertrophic cardiomyopathy that have a low prevalence, are underdiagnosed, and lack specific remarkable features on ECGs.[95] Future studies will need to elucidate the degree to which such anatomical defects can be replicated in the model. In addition, generative models trained on one data modality can create realistic missing data. For example, CMRs have been generated from ECGs using an autoencoder trained with a small dataset of paired CMRs and ECGs (*Figure 3*).[96] Moreover, when MRI and ECG data are available, information on genomic variants can also be inferred.[96] The need for training data can also be reduced by enhancing datasets with generative models; however, whether the generated data has any useful new information is uncertain. Electrocardiography models pre-trained using a transformer model detected left ventricular dysfunction and hypertrophic cardiomyopathy with fewer examples, though their performance has been outperformed by other models trained directly with more examples.[90,91,97–99]

Digital twins for data generation: clinical trial efficiency and translation

Clinical trial efficiency

Beyond optimizing individual care, digital twins enhance the efficiency of the evidence-generation process itself (*Figure 5*). Randomized clinical trials represent the backbone of evidence-based practice but are challenged by the time and resources it takes to conduct them. Digital twins can enhance the efficiency of RCTs, such as the augmentation of the control arm based on the relevant patient population in the EHR. By enrolling fewer controls, digital twins could, therefore, reduce recruitment time and cost.

Proof-of-concept studies of control arm augmentation with digital twins have demonstrated survival curves similar to the original participants, though some demographic and clinical value distributions differ between populations,[100,101] suggesting the reliability of inference from these experiments. With observational cohort data, several studies have used GANs for feature and participant balance across the cohorts to improve treatment effect estimates.[84,85,102,103] Finally, the entire trial enterprise will likely undergo significant upheaval with several applications of statistical machine learning and genomically enabled trial enrichment poised to transform how evidence is generated.[104,105]

Clinical trial translation

Digital twins of patient cohorts can also assess the generalizability of treatment effects in RCTs, particularly by projecting these effects in real-world populations (*Figure 5*). Randomized clinical trials estimate the average treatment effect of an intervention across all included individuals. However, there is often under-enrolment of key populations

such as women, minorities, older adults, and patients with multiple comorbidities.[106] Moreover, there is demonstrable heterogeneity of treatment effects in many trials.[56,57,107] To translate evidence from one patient population to another, digital twins can simulate multiple patient population characteristics by constructing a synthetic cohort that accurately represents their covariate distributions. They utilize GANs, which simulate treatment effects in these synthetic patient populations, to reconcile discrepancies in treatment effects between different established patient populations.[108] This has been demonstrated through studying a pair of RCTs with the same intervention and primary outcome, SPRINT and ACCORD, in which a SPRINT-conditioned ACCORD digital twin reproduced the SPRINT average treatment effect and vice versa.[79]

Challenges and ethical considerations

Challenges

Digital twins of medical data have special challenges and ethical considerations. These span the special privacy controls, fairness of algorithms, trust, reliability, and durability of the model predictions, and the infrastructure and cost required in developing these twins. The next section outlines approaches to these challenges to ensure responsible deployment in healthcare settings.[109]

Infrastructure and technological challenges

The digital twin infrastructure houses multi-modal data, stores and continually updates patient sensor data, and utilizes high-dimensional processing power to integrate the data and update prediction models. Storage and computing sources are expensive, but in healthcare settings, they also require compliance with patient security laws and provisions and adequate testing against attacks attempting to reverse-engineer source data.[110]

Acquiring the right data for the digital twins can also be challenging. Electronic health record data are sparse; often, these data are not missing at random, and their absence can be clinically informative. Prospective cohorts or registries such as the UK Biobank have a comprehensive diagnostic repository with data uniformly recorded in 'well' humans but are limited by a lack of global representation and low rates of pathology and outcomes.[74] In addition, 'omics' data are not commonly acquired in hospitals or clinics, limiting their generalizability across digital twins. Fitness trackers rely on consumer uptake and constant use. Representative studies have shown, e.g., that patients who would benefit from wearable technology the most, such as those with cardiovascular disease, often do not use the technology.[111,112]

Multi-modal data vary across formats, frequency of capture, and storage, requiring a concerted effort to promote and build the infrastructure for standardized data organization and interoperability across healthcare devices and software.[113] Strategies to address these challenges include broader compliance with interoperable data formats, such as DICOM,[114] and standardized structure for EHR research across institutions through HL7 fast healthcare interoperability resources (FHIR) and the Observational Medical Outcomes Partnership Common Data Model (OMOP CDM).[115,116] Technical solutions that leverage processing disparate data with autoencoders or formatting images to resemble other imaging modalities are examples of successful integration of multi-modal data into AI models in cardiology.[49,96]

Ensuring data privacy

Digital twins in cardiovascular medicine use sensitive health information drawn from the EHR, genomic and other diagnostic testing, and continuously connected wearable devices, requiring special precautions for storage and access to prevent re-identification.[117] The models must be evaluated frequently with differential privacy measures for any leakage of the training data.[118] Digital twins that use decoder sampling or GAN models preserve covariate distributions and correlations between covariates of the real data without revealing original information.[119–121] The model's output can also be analysed to assess whether the generated data reveals specific information about individual training samples using techniques such as membership inference attacks.[122] Employing anonymization methods such as data perturbation or tokenization also reduces the risk of the re-identification of individuals in the training data. Regular privacy impact assessments and audits throughout the model development lifecycle help identify and address potential privacy vulnerabilities.

The need for digital twin outcome data

Digital twins simulate many permutations of an intervention on a patient, but the efficacy of this intervention on patient health outcomes, along with interventions powered by AI, is only starting to be tested in healthcare settings. In particular, the single known RCT is evaluating Type 2 diabetes remission through digital twin monitoring.[123] Depending on trial feasibility and clinical risk, testing for digital twin efficacy can vary for regulatory approvals for medical devices, including those with robust modelling results but no clinical testing.[124] Without clinical validation, however, the accuracy and potential impacts on patient safety for predictions and treatment recommendations from digital twins are unknown.[125] Therefore, RCTs are needed to establish the efficacy of clinical paradigms that leverage digital twins before their clinical implementation.

Ethical considerations

Fairness

In patient populations, under-representation and selection biases can impact the usability of synthetic datasets for downstream tasks. For example, EHR cohorts have high volume health data but are limited by biases in patient selection, data acquisition and documentation, and sparseness since the data is collected during healthcare encounters.[126] Therefore, to evaluate whether generated data perpetuates biases present in the training set, particularly concerning sensitive attributes like race or gender, fairness metrics such as demographic parity and equalized odds can be implemented.[127] Furthermore, diversity metrics can assess the variety and representativeness of the generated data across different demographic groups. Despite continued technical developments, bias correction for generative deep learning has been challenging to overcome. Recent developments align the consistency of the distribution of inputs and outputs[128] with the intent to make the digital twins more reliable.

Disparities in healthcare and technology access can limit the information available for many populations, leading to suboptimal representation in digital twin model training. For instance, older adults, minorities, and individuals with low income use wearable devices as well as smart devices at a lower rate than their privileged counterparts.[112,129] Moreover, in deployment, differential access to high bandwidth internet connectivity can further hamper equitable use. Digital

twin development and deployment must, therefore, ensure the inclusion of safety-net and rural hospitals to avoid the widening of health disparities. This could be through many strategies, including incentivizing the reimbursement of models that are generalizable across health systems and demographic groups.[130]

Reliability and trust

Digital twins can integrate a vast amount of data and provide intervention recommendations, but the predictions must be reliable and trustworthy for clinicians to act upon them. Artificial intelligence algorithms historically are not explainable, making it difficult for physicians to understand or trust their output. Current methods such as Shapley Additive Explanations and Gradient-weighted Class Activation Mapping, which explain and visualize feature importance in deep learning models, enable a degree of explainability to black box AI models.[131,132] However, more sophisticated explainability methods are essential to broaden trust. Moreover, patients must be fully informed of the privacy consequences of real-time health monitoring with potential interventions outside of a healthcare setting before consent is obtained to participate in care strategies that leverage digital twins.[7] The AI Trust, Risk, and Security Management (TRiSM) framework has standardized how to address central ethical, practical, and implementable tenants of AI models.[133]

In addition, the accuracy of prediction models deployed in healthcare drifts over time, paradoxically, from constant retraining with new patients.[134] The changes in the diagnostic and therapeutic landscape may contribute to this drift, along with false correlations between covariates and outcomes from past patients. This reiterates the need for more outcomes research with digital twin implementation.

Data ownership

Since the digital twins are created from patient data, ethics of data ownership, especially after a patient's death, must be addressed.[8] The European Union General Data Protection Regulation legislation and the US Health Insurance Portability and Accountability Act give citizens control of their data, but their application to digital twins is not established.[135–137] An ethical framework must be developed to address data ownership, utilization, and patient autonomy.

Considerations for implementation at scale

Government regulation

In the USA, AI tools in healthcare, such as digital twins, are regulated by the Food and Drug Administration (FDA) as a Software as a Medical Device. They are classified based on the severity of the medical condition and the extent to which the software contributes to medical management.[138] As digital twin models become embedded in healthcare, their regulation will establish new privacy standards for continuous and real-time syncing of patient wearables to data warehouses, developing patient autonomy over their digital twin, and promoting trust and safety in the models.[139] The AI Act from the European Union has added measures to reduce the personal risks of AI and ensure constant re-evaluation and maintenance of the models.

Building international collaboration and federated learning

For regulatory clearance and subsequent use of digital wins, broad validation networks are needed. The charge for developers of models for any healthcare application, especially for digital twins, is that they reliably simulate scenarios across varying data availability, processing pipelines, healthcare systems, and patient populations. International collaborations utilize standardized data organization, such as the OMOP CDM, which is implemented by institutions and major data registries globally as a standard set of relational tables to map EHR data, allowing for multi-national characterizations of hundreds of millions of patients.[140–142] Data-sharing consortiums such as the EUDAT Collaborative Data Infrastructure and the Information Commons established by the US 21st Century Cures Act are important steps in building the foundation for federated learning.[135,143] UNESCO has also established an open data policy which emphasizes interoperability and accessibility through sharing of data resources.[144] Federated learning bypasses data transfer and potential leakage across institutions by decentralizing model training and testing to the data site, and digital twins built from a distribution can prevent original data from being shared (*Figure 5*).[110]

Implementation in healthcare systems

Once clinically relevant and accurate models have been successfully developed and tested through RCTs, they will need to be implemented within healthcare systems. This will require the creation and maintenance of the infrastructure to store, organize, and process multi-modal data that are input to these models and to refine the models across diverse patient populations. In addition, clinicians will need to adjust their workflow and be trained on how to evaluate and act on recommendations from digital twin models.

Moreover, reimbursement models will need to be structured to align with the resources necessary to develop these models and the value they provide. Currently, there are no strategies for reimbursing digital twin-enabled care, so hospitals must invest in these emerging technologies. In the USA, the Center for Medicare and Medicaid Services (CMS) has reimbursed certain single-purpose FDA-cleared software, spanning AI diagnostics and assisted triage systems.[130] European countries have varying reimbursement policies for digital health, but none for AI models.[145] These predate the current movement towards comprehensive foundation and digital twin models.[13] However, given the shift to value-based care in the USA and in nationalized health systems, the efficiency gained by physicians through increased patient volumes and better outcomes may factor into financial models that can support the broader use of digital twin technology.[130]

Future directions

The future of digital twins in cardiovascular care will move beyond the traditional static 3D models of patient physiology into expansive tools examining healthcare data in real-time and enabling the selection and deployment of personalized interventions to prevent adverse cardiovascular outcomes. The fusion of multi-modal medical data, advanced deep learning models, and generative AI will bring cardiology twins closer to effectively and bidirectionally transferring complex processes from the physical world to the *in silico* one. Through comprehensive registries, data from all health encounters in EHRs, and personalized health data available from wearable devices, digital twin models will have access to millions of examples to develop accurate predictions

on individual patients. Despite these advances, digital twins will need to be pragmatic with the data available, clinicians will need to understand the uncertainties behind probabilistic recommendations, and models will need to be continuously re-evaluated. Prospective trials of the models with multicenter and multi-national validation will ensure the feasibility and generalizability of digital twin deployment, with the need for special emphasis on fairness. The tools for building a digital twin, complete with physiological, environmental, and systemic data to accelerate cardiovascular care, are available. The barriers to entry are surmountable with a co-ordinated, multi-national effort to prioritize data and model sharing to advance precision cardiovascular care through digital twins.

Declarations

Disclosure of Interest

Dr Thangaraj, Dr Oikonomou, and Dr Khera are coinventors of a provisional patent related to the current work (63/606,203). Dr Khera is an Associate Editor of JAMA and receives research support, through Yale, from the National Heart, Lung, and Blood Institute of the National Institutes of Health (R01HL167858 and K23HL153775), the Doris Duke Charitable Foundation (2022060), the Blavatnik Foundation, Bristol-Myers Squibb, Novo Nordisk, and BridgeBio. He is a co-inventor of other pending patents, WO2023230345A1, US20220336048A1, 63/346,610, 63/484,426, 63/508,315, 63/580,137, 63/619,241, and 63/562,335, and a co-founder of Ensight-AI, Inc. and Evidence2Health, LLC. Dr Oikonomou is an academic co-founder of Evidence2Health LLC, and has been a consultant for Caristo Diagnostics, Ltd and Ensight-AI, Inc. He is a co-inventor in patent applications (US17/720,068, 63/619,241, 63/177,117, 63/580,137, 63/606,203, 63/562,335, WO2018078395A1, WO2020058713A1) and has received royalty fees from technology licensed through the University of Oxford. Dr Asselbergs is supported by Heart4Data, which received funding from the Dutch Heart Foundation and ZonMw (2021-B015), and UCL Hospitals NIHR Biomedical Research Centre.

Data Availability

No data were generated or analysed for or in support of this paper.

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